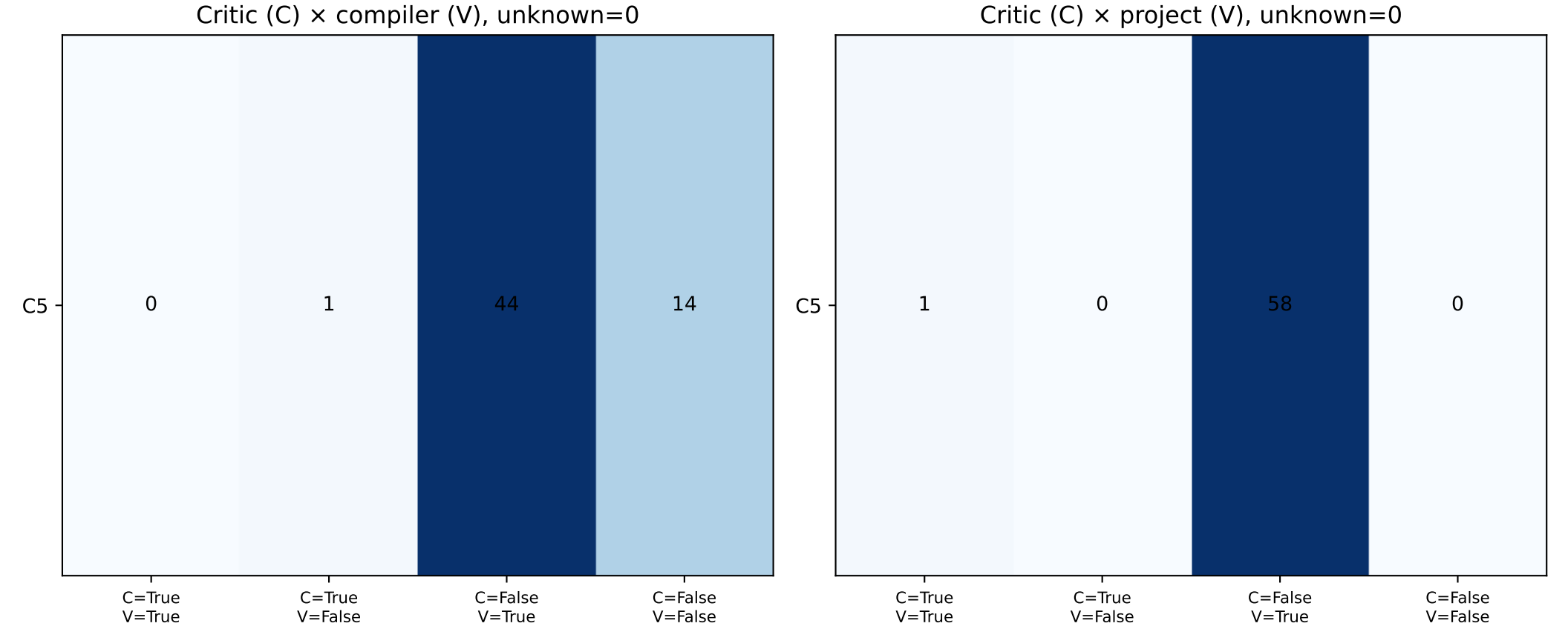Observed case-run counts; repeated cases are not independent samples